Stationary Strings Theorem V: Arrest of the September 2026 Navier–Stokes Vortex Singularity by Elastic String Tension, $X \to 0$ Axis Regularization and Self-Consistent Saturation Proof

(Analytical Derivation in the (X, η, q) Similarity Coordinates of the Millennium Vortex Solution, Cartesian Axis Regularization and High-Precision Radau/DOP853 Numerical Simulation Results)

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Abstract

The September 2026 study *Finite Time Blowup for Navier–Stokes* [1] proves that a smooth fluid starting from rest in the three-dimensional incompressible Navier–Stokes equations develops, in finite time ($t \uparrow 1$, $\tau = 1 - t \to 0$), an axisymmetric vortex core reaching infinite velocity ($\|u_\theta^{(0)}\|_{L^\infty} \asymp \tau^{-1/2-h} \to \infty$). The singularity is driven, in similarity coordinates (X, η, q), by the radial pressure drop $\Pi_X^{(0)} = F(X, \eta)^2$ produced by centripetal acceleration and by the cylindrical shear stress $\Sigma_{r\theta} = -q^{-1-h} F(X, \eta) a(X, \eta)$ [1, pp. 8, 26, 27].

In this work, the "dimensional shear and singularity-free core" mechanism predicted by the *Stationary Strings Theorem* [2–5] is applied directly to the September 2026 Navier–Stokes inner-core equations, the $\sim 1/X$ coordinate singularity on the $X \to 0$ symmetry axis is removed by a Cartesian regularity factor ($\ell_\theta = \sqrt{X/X_*}\,\tilde{\ell}$), and the resulting coupled differential system is solved in the parameter space $h \in \{0.002, 0.005, 0.008, 0.010\}$ and $\mu_s \in \{0.02, 0.05, 0.10, 0.20\}$ with high-precision solvers (`Radau`, `DOP853`, `brentq`, 10^{-14} tolerance):

1. **Definite Singularity Arrest in the Explicit Model:** Under the negative hoop tension $-\frac{T_{\theta\theta}^{\mathrm{sicim}}}{r}$ produced by one-dimensional elastic strings, the effective pressure gradient vanishes at $s_{\mathrm{kritik}} = -\ln \tau_{\mathrm{kritik}} \in [16.1196793715, 58.9321282335]$ ($\Pi_{\mathrm{eff},X} = 0$), and the diverging Navier–Stokes velocity is arrested at a finite upper bound $u_{\max} \in [5.92 \times 10^2, 1.02 \times 10^{12}]$.

2. **Self-Quenched Saturation in the Fully Self-Consistent Model:** When the nonlinear backreaction of the string tension on the effective shear (F_{eff}) is included, the system is found to choke itself (*self-quenching*) and reach a stable asymptotic saturation at $\Pi_{\text{eff}} \approx 0.181 - 0.187 > 0$ ($s = 40$) instead of a sharp zero crossing.

3. $X \to 0$ **Axis Regularization (Theorem 2 and Test T6):** With the transformation $\ell_\theta(X, s) = \sqrt{X/X_*}\,\tilde{\ell}(X, s)$ and the driving $G(X) = F\sqrt{a^2 + \beta_0^2 l^2}$ ($\beta_0 = 0.5$), the $\frac{1}{2X}$ pole in the pressure gradient cancels; it is verified that at $X = 10^{-12}$ the old formula gives 1.68×10^6 while the regularized formula gives 0.0478 (contrast $\sim 3.5 \times 10^7$), with $\Pi_{\text{reg}}(0) = 0.952236996205853$ on the exact axis and a finite inner-core central pressure integral $\Pi_{\text{eff,reg}}(0, s) = -0.876471225586346$.

4. **Acoustic Event Horizon and Topological Information Invariant:** Under the radial inflow profile $V_0(X, \eta)$, the wave frequency is verified to vanish ($\omega' \to 0$) at the scale $q_{\text{ufuk}}(X, \eta) \in [9.76 \times 10^{-4}, 1.20 \times 10^{-2}]$, and the topological information invariant is verified to be conserved at machine precision, $\max|\mathcal{I}_n - nc_s| \le 7.11 \times 10^{-15}$, for the string modes $n \in \{1, 2, 3, 5\}$.

5. **Cutoff-Free Global Convergence on $[0, \infty)$ and Proposition 4.2 Stress Quenching (Theorem 3 / Tests T7–T8):** In the balance-subtracted ($\tilde{\ell}^2 - \ell_0^2$) global model with balance-preserving relaxation, the far-field tail is shown to decay as $O(X^{-1-h})$ and the infinite $[0, \infty)$ central pressure integral to converge with $\sim 10^{-14}$ `quad` error without any cutoff; moreover, under the Proposition 4.2 transport operators of [1, pp. 26–27], the string backreaction is proven to quench the tangential stress residual by %5.31 in the $\|\mathcal{T}_0\|_{L^2}$ norm and by %14.34 in the $\|\mathcal{T}_0\|_{L^\infty}$ norm.

6. **Full (X, s) FENE-P PDE Solution and Rational Curvature Factor (Theorem 4 / Tests T9–T10):** In the 2D PDE solution including radial advection $W(X, 0)X\partial_X\tilde{\ell}$ and the FENE-P finite-extensibility bound ($b_{\max} = 10^{16}$), core saturation ($\max\tilde{\ell}^2/b_{\max} = 0.9943 < 1$) and finite velocity ($u_{\max} \approx 3.83$–$4.00 \times 10^8$) are shown to persist; under the rational cylindrical curvature factor $\mathcal{K}_{\text{curv}}(X) = \frac{X_0 + X_*}{2X_*(X_0 + X)}$, the PDE far-field wake exponent rises from $q = 0.9098$ to $q = 1.9010 > 1$, so that the global PDE integral ($I_\infty \approx 0.867336$) also converges without cutoff.

7. **Two-Field Dynamic PDE and Skew-Symmetric Lyapunov Power Cancellation (Theorem 5 / Test T11):** Raising the fluid velocity from an algebraic to a fully dynamic field, with deviation advection $W[X\partial_X(F - F_{\text{base}}) + (1 + h)(F - F_{\text{base}})]$ and F-proportional skew-symmetric FENE-P coupling, the reactive powers cancel pointwise, $P_{\text{fluid}} + P_{\text{string}} \equiv 0$ (error $\sim 6 \times 10^{-16}$); positivity ($0 < F \le F_{\text{base}}$, bitwise base preservation at $\mu_s = 0$) and convergent $I_\infty \approx 0.903$–0.951 with $q \approx 2.00$ are preserved; the core velocity is suppressed to $F(X_*, 40) \approx 0.483$–$0.491 < 0.498$.

8. **n-Mode Spectral Transfer in the Co-Moving Lagrange Frame (Test T12):** On a uniform $N_\xi = 128$ label grid (per-parcel ODEs, no upwind diffusion and no convolution kernel), the $n = 1, 2, 3, 4$ quantum modes are verified to transfer into the fluid velocity field with exact 4/4 $k_{\max} = n$ matching and %90.6–%99.2 spectral purity ($|r| = 0.8143$–0.9453); the Euler-window $[\,0.004, 0.040\,]$ escape is recorded as the justification of the Lagrange frame.

1 Introduction and Problem Statement

In the first four studies of the *Stationary Strings Theorem* [2–5], it was theoretically modeled that matter entering vortex singularities splits into two opposite dimensions, **String (Frequency)** and **Light (Flow)**: while the extrinsic flow dimension stretches to infinity ($\mathcal{L} \to \infty$), the intrinsic oscillation frequency collapses to zero ($\omega_n \to 0$), and the negative-pressure string core forming at the center removes the singularity.

The 166-page September 2026 study *Finite Time Blowup for Navier–Stokes* [1] proves the formation of a velocity singularity $\limsup_{t\uparrow 1} \|u(t)\|_{L^\infty(\mathbb{R}^3)} = \infty$ under finite kinetic energy in the three-dimensional incompressible Navier–Stokes equations ($\partial_t u + (u\cdot\nabla)u - \nu\Delta u + \nabla p = f,\ \nabla\cdot u = 0$) [1, Theorem 1.1, p. 1]. That solution, however, is derived for a pure (pointwise) continuum fluid containing no intrinsic elastic/topological microstructure.

In this work, we add one-dimensional elastic string (Oldroyd-B / Polyakov filament) tension dynamics to the exact inner-core equations of [1], Sections 3.1 (pp. 7–8) and 4.1 (pp. 25–27), deriving both the analytical theorems and the $X \to 0$ axis regularization, and present the verification results (`Test T1-T12`) obtained with the `duran_sicimler_sim` numerical solver package.

2 September 2026 Navier–Stokes Inner-Core Equations and Analytical Reference Profiles

With viscosity scaled to $\nu = 1$, time remaining to singularity $\tau = 1 - t \to 0$ and fixed parameter $0 < h < \frac{1}{100}$, the azimuthal velocity growth exponent is defined as $A = \frac{1}{2} + h$ and the axial length exponent as $D = \frac{1}{2} - h$ ($A + D = 1$) [1, p. 7]. In cylindrical coordinates (r, θ, z), with concentration scale $q = q(z, \tau) > 0$, the dimensionless similarity coordinates (X, η) are [1, Equation (3.2), p. 7; p. 25]:

$$\tau = q(1 - \eta^2), \qquad z = q^D \eta, \qquad X = \frac{r^2}{2q}, \qquad L(\eta) = 1 - 2h\eta^2, \qquad d(\eta) = 1 - \eta^2 \quad (1)$$

The leading velocity components and pressure of the vortex core are [1, Equations

(4.3)–(4.4), p. 25]:

$$\begin{aligned} u_\theta^{(0)} &= q^{-A}E(X,\eta), \qquad u_z^{(0)} = q^{-A}U(X,\eta), \\ ru_r^{(0)} &= V_0(X,\eta), \qquad p^{(0)} = q^{-2A}\Pi^{(0)}(X,\eta) \end{aligned} \tag{2}$$

with the on-axis ($X = 0$) Cartesian regularity condition $E(X,\eta) = \sqrt{2X}F(X,\eta)$. Incompressibility and the centripetal-pressure balance (`navier-stokes.pdf` p. 26, **Equation 4.7**) give the fundamental relations [1, p. 26]:

$$\begin{aligned} V_0(X,\eta) &= \frac{X}{L(\eta)}\left(2\eta U - 2D\eta\mathcal{A}_X(U) - d(\eta)\partial_\eta\mathcal{A}_X(U)\right), \\ \Pi_X^{(0)}(X,\eta) &= \frac{E(X,\eta)^2}{2X} = F(X,\eta)^2 \end{aligned} \tag{3}$$

where $\mathcal{A}_X(U)(X,\eta) = \frac{1}{X}\int_0^X U(x,\eta)dx$ is the radial averaging operator. Per **Equations (4.8) and (4.11)** on pp. 26–27 [1, pp. 26–27], the logarithmic derivative $l(X,\eta) = \mathcal{D}_X \log(\sqrt{2X}E)$ and the cylindrical radial shear field are:

$$\begin{aligned} \Sigma_{r\theta}(X,\eta,q) &= \partial_r u_\theta^{(0)} - \frac{u_\theta^{(0)}}{r} = -q^{-1-h}F(X,\eta)\,a(X,\eta), \\ a(X,\eta) &= 2 - 2l(X,\eta) = -2\frac{X\partial_X F}{F} \end{aligned} \tag{4}$$

Our solver uses analytical reference profiles satisfying both the $X \to 0$ on-axis regularity ($E/\sqrt{2X} \to F_0$) and the $X \to \infty$ asymptotic decay ($E \sim c_\infty X^{-1/2-h}$) of [1, p. 8] exactly:

$$\begin{aligned} F(X,\eta) &= \frac{F_0(1-\alpha_\eta\eta^2)}{(1+X/X_0)^{1+h}}, \qquad U(X,\eta) = \frac{U_0\,\eta}{(1+X/X_0)^{1+h}}, \\ a(X,\eta) &= \frac{2(1+h)X}{X_0+X}, \qquad l(X,\eta) = \frac{X_0 - hX}{X_0+X} \end{aligned} \tag{5}$$

For this profile the algebraic identity $l(X,0) = 1 - \frac{1}{2}a(X,0)$ holds exactly (numerical verification error 1.11×10^{-16}).

3 Elastic String Hoop Tension ($-\frac{T_{\theta\theta}}{r}$), Singularity Arrest and $X \to 0$ Axis Regularization

3.1 String Evolution and Arrest on the Reference Core Ring ($X = X_*$, Theorem 1)

Defining the logarithmic singularity time $s = -\ln q = -\ln \tau \in [0, \infty)$ ($q = e^{-s}$), the azimuthal elongation component $\ell(s) \equiv \ell_\theta(X_*, s)$ of the one-dimensional elastic string on the vortex core ring $X_* > 0$ evolves under the $\Sigma_{r\theta}$ shear of Equation (4) and the $\gamma_{\rm rel}$ relaxation term as:

$$\frac{d\ell(s)}{ds} = \kappa_r \, F_{\rm eff}(X_*, s) |a(X_*, 0)| \, e^{hs} \, \ell(s) - \gamma_{\rm rel} \, e^{-s} \, \ell(s), \qquad \ell(0) = \ell_0 \tag{6}$$

The negative hoop tension $-\frac{T^{\rm sicim}_{\theta\theta}}{r}$ produced by the stretched string ring in cylindrical coordinates ($T^{\rm sicim}_{\theta\theta} = \mu_s \ell(s)^2$) modifies the radial pressure gradient of Equation (3) and the effective velocity profile with feedback as:

$$\begin{aligned} \Pi_{{\rm eff},X}(X_*, s) &= F(X_*, 0)^2 - \frac{\mu_s \, e^{-(1+2h)s} \, \ell(s)^2}{2X_*}, \\ u^{\rm eff}_\theta(X_*, s) &= e^{(1/2+h)s} \sqrt{\max\left(0, \, 2X_* \Pi_{{\rm eff},X}(X_*, s)\right)} \end{aligned} \tag{7}$$

In the explicit model neglecting backreaction ($F_{\rm eff} = F(X_*, 0)$), Equation (6) is solved exactly in closed form:

$$\ell_{\rm expl}(s) = \ell_0 \exp\left(\frac{\kappa_r F(X_*, 0) |a(X_*, 0)|}{h} \left(e^{hs} - 1\right) - \gamma_{\rm rel} \left(1 - e^{-s}\right) \right) \tag{8}$$

While $\ell_{\rm expl}(s)^2$ grows at a double-exponential (stretched-exponential) rate, the factor $e^{-(1+2h)s}$ decays only exponentially; hence for every $\mu_s > 0$ and $h > 0$ there exists a finite $s_{\rm kritik} < \infty$ ($\tau_{\rm kritik} = e^{-s_{\rm kritik}} > 0$) with $\Pi_{{\rm eff},X}(X_*, s_{\rm kritik}) = 0$, and the velocity is arrested at $u_{\max} = \max_{s \in [0, s_{\rm kritik}]} u^{\rm eff}_\theta(X_*, s) < \infty$.

3.2 Theorem 2: $X \to 0$ Cartesian Axis Regularization (Axis-Regularization)

If the elongation $\ell(X, s)$ in Equation (7) is kept constant ($\ell_0 > 0$) in the $X \to 0$ limit, the term $-\frac{\mu_s \ell^2}{2X}$ produces a $\sim 1/X$ pole on the $X \to 0$ symmetry axis.

Theorem 2 (Cartesian Axis Regularization): *By cylindrical on-axis regularity (just as in the $E(X, \eta) = \sqrt{2X} F(X, \eta)$ [1, Equation (4.4)] structure), the azimuthal string elongation is defined as:*

$$\ell_\theta(X,s) = \sqrt{\frac{X}{X_*}}\,\tilde{\ell}(X,s), \qquad \tilde{\ell}(X,0) = \ell_0 \tag{9}$$

Inserting Equation (9) into the original feedback term $\frac{\mu_s e^{-(1+2h)s}\ell_\theta^2}{2X}$ *cancels the* $2X$ *factors identically over the entire* $X \in [0,\infty)$ *domain:*

$$\Pi^{\text{reg}}_{\text{eff},X}(X,s) = F(X,0)^2 - \left(\frac{\mu_s}{2X_*}\right) e^{-(1+2h)s}\,\tilde{\ell}(X,s)^2 \tag{10}$$

and the equation contains no $1/X$ *singularity. The reduced string amplitude* $\tilde{\ell}(X,s)$ *is driven by the core strain rate* $G(X)$*, combining both the differential shear* $a(X,0)$ *and the core rotation/compression invariant* $l(X,0) = \frac{X_0 - hX}{X_0 + X}$*:*

$$\begin{aligned} \frac{d\tilde{\ell}(X,s)}{ds} &= \kappa_r\, G(X)\, e^{hs}\,\tilde{\ell}(X,s) - \gamma_{\text{rel}}\, e^{-s}\,\tilde{\ell}(X,s), \\ G(X) &= F(X,0)\sqrt{a(X,0)^2 + \beta_0^2\, l(X,0)^2} \end{aligned} \tag{11}$$

On the $X = 0$ *axis,* $a(0,0) = 0$ *and* $l(0,0) = 1$*, so* $G(0) = F_0\beta_0 > 0$ *is exactly finite; the zero-driving problem on the axis* $(F|a|(0) = 0)$ *thus disappears,* $\Pi^{\text{reg}}_{\text{eff},X}(0,s)$ *stays finite on the exact axis, and the cumulative central pressure integral* $\Pi_{\text{eff,reg}}(0,s) = -\int_0^{X_{\max}} \Pi^{\text{reg}}_{\text{eff},X}(X,s)\,dX$ *over the inner-core domain* $[0, X_{\max}]$ *is absolutely finite.* □

3.3 Theorem 3: Balance-Subtracted Global Model and Cutoff-Free Convergence on $[0,\infty)$ (Test T7)

In Oldroyd-B and Polyakov string mechanics the base equilibrium length ℓ_0 produces no net force; the net elastic hoop tension acting on the fluid is $\Delta T_{\theta\theta} = \mu_s(\ell_\theta^2(X,s) - \ell_\theta^2(X,0))$, i.e. the difference $(\tilde{\ell}(X,s)^2 - \ell_0^2)$ in the reduced amplitude. Writing the relaxation term in the **balance-preserving** form $-\gamma_{\text{rel}} e^{-s}(\tilde{\ell} - \ell_0)$ gives:

$$\frac{d\tilde{\ell}(X,s)}{ds} = \kappa_r\, G(X)\, e^{hs}\,\tilde{\ell}(X,s) - \gamma_{\text{rel}}\, e^{-s}\left(\tilde{\ell}(X,s) - \ell_0\right), \qquad \tilde{\ell}(X,0) = \ell_0 \tag{12}$$

$$\Pi^{\text{global}}_{\text{eff},X}(X,s) = F(X,0)^2 - \left(\frac{\mu_s}{2X_*}\right) e^{-(1+2h)s}\left(\tilde{\ell}(X,s)^2 - \ell_0^2\right) \tag{13}$$

Theorem 3: *Since in the far field* $(X \to \infty)$ $G(X) \sim C_G X^{-1-h}$ $(C_G = F_0\sqrt{4(1+h)^2 + \beta_0^2 h^2})$ *vanishes, and since balance-preserving relaxation preserves the* $\tilde{\ell} = \ell_0$ *equilibrium in undriven regions,* $(\tilde{\ell}(X,s)^2 - \ell_0^2) \sim 2\ell_0^2\kappa_r K(s) G(X) \sim O(X^{-1-h})$ *decays. Hence* $\Pi^{\text{global}}_{\text{eff},X}(X,s)$ *is both pole-free on the* $X = 0$ *axis and absolutely integrable over the infinite* $[0,\infty)$ *interval with no cutoff (*$\left|\int_0^\infty \Pi^{\text{global}}_{\text{eff},X}(X,s)dX\right| < \infty$*).* □

3.4 Proposition 4.2 Transport Operators and Tangential Stress Residual $\mathcal{T}_{0,r\theta}$ (Test T8)

Per `navier-stokes.pdf` p. 8 (Proposition B.2) and pp. 26–27 (**Proposition 4.2, Equations 4.8–4.11**), the general axial velocity profile with nonzero even component on the axis ($U_{\text{even}} = 0.5$) and the transport operators are:

$$U_{\text{full}}(X,\eta) = \frac{U_{\text{even}}(1-\alpha_\eta\eta^2) + U_0\,\eta}{(1+X/X_0)^{1+h}} \tag{14}$$

$$\begin{cases} W(X,\eta) = 1 - 2D\eta\mathcal{A}_X(U_{\text{full}}) - d(\eta)\partial_\eta\mathcal{A}_X(U_{\text{full}}) \\ H_c(X,\eta) = D\eta + d(\eta)U_{\text{full}}(X,\eta) \end{cases} \tag{15}$$

The inviscid angular momentum source $S_q(X,\eta)$, the on-axis-regular ($C_Q = 0$) radial integral $Q_s(X,\eta)$ and the tangential stress residual $\mathcal{T}_{0,r\theta}(X,\eta)$ [1, Equations (4.9)–(4.11), pp. 26–27]:

$$\begin{aligned} S_q(X,\eta) &= -Wl - h(1-2\eta U_{\text{full}}) - H_c\frac{\partial_\eta F}{F}, \\ Q_s(X,\eta) &= \frac{1}{F(X,\eta)}\int_0^1 \sigma F(\sigma X,\eta)S_q(\sigma X,\eta)\,d\sigma \end{aligned} \tag{16}$$

$$\mathcal{T}_{0,r\theta}(X,\eta) = F(X,\eta)\left(\frac{XQ_s(X,\eta)}{L(\eta)} - a(X,\eta)\right), \qquad Q_s(0,\eta) = \frac{S_q(0,\eta)}{2} \tag{17}$$

take the above form. On the midplane ($\eta = 0, X = 0$), $W(0,0) = 0, l(0,0) = 1, \partial_\eta F(0,0) = 0$, so the analytic identities $S_q(0,0) = -h$ and $Q_s(0,0) = -h/2$ hold exactly.

3.5 Full (X,s) Advective PDE + FENE-P Model (Test T9)

Adding the radial advection operator $+W(X,0)X\partial_X\tilde{\ell}$ of Equation (4.14) on p. 27 of `navier-stokes.pdf` and the FENE-P finite-extensibility bound ($b_{\max} = \ell_{\max}^2$ (10^{16})), the full partial differential equation (PDE) system in (X,s) space reads:

$$\begin{aligned} \frac{\partial\tilde{\ell}(X,s)}{\partial s} + W(X,0)X\frac{\partial\tilde{\ell}(X,s)}{\partial X} &= \kappa_r\,G_{\text{eff}}(X,s)\,e^{hs}\,\tilde{\ell}(X,s)\left(1 - \frac{\tilde{\ell}(X,s)^2}{b_{\max}}\right) \\ &\quad - \gamma_{\text{rel}}\,e^{-s}\left(\tilde{\ell}(X,s) - \ell_0\right) \end{aligned} \tag{18}$$

$$\Pi^{\mathrm{PDE}}_{\mathrm{eff},X}(X,s) = F(X,0)^2 - \left(\frac{\mu_s}{2X_*}\right) e^{-(1+2h)s} \frac{\tilde{\ell}(X,s)^2 - \ell_0^2}{1 - \tilde{\ell}(X,s)^2/b_{\max}},$$
$$F_{\mathrm{eff}}(X,s) = \sqrt{\max\left(0,\, \Pi^{\mathrm{PDE}}_{\mathrm{eff},X}(X,s)\right)} \tag{19}$$

3.6 Theorem 4: Rational Cylindrical Curvature Factor and Global PDE Convergence (Test T10)

In the advective 2D PDE model, positive radial advection ($dX/ds = W(X,0)X > 0$) carries the FENE-saturated core strain to the outer region ($X \to \infty$), forming a slowly decaying wake with exponent $q_{\mathrm{wake}} \approx 0.9098 < 1$. Under the constant $\frac{1}{2X_*}$ coefficient this wake diverges the $[0,\infty)$ integral, because the $\ell_\theta = \sqrt{X/X_*}\,\tilde{\ell}$ transformation removes the physical $\frac{1}{2X}$ cylindrical curvature damping at the $X \to \infty$ boundary.

Theorem 4 (Rational Curvature Factor): *When the tangential string elongation is defined with a Padé-type rational factor preserving both Cartesian regularity on the $X \to 0$ axis ($\ell_\theta \sim \sqrt{X}$) and physical finite extensibility in the $X \to \infty$ far field ($\ell_\theta \sim \tilde{\ell}$):*

$$\ell_\theta(X,s) = \sqrt{\frac{(X_0 + X_*)X}{X_*(X_0 + X)}}\,\tilde{\ell}(X,s),$$
$$\mathcal{K}_{\mathrm{curv}}(X) = \frac{\ell_\theta(X,s)^2}{2X\,\tilde{\ell}(X,s)^2} = \frac{X_0 + X_*}{2X_*(X_0 + X)} \tag{20}$$

the effective PDE pressure gradient takes the following fully global form:

$$\Pi^{\mathrm{PDE,curv}}_{\mathrm{eff},X}(X,s) = F(X,0)^2 - \mathcal{K}_{\mathrm{curv}}(X)\,\mu_s\, e^{-(1+2h)s} \frac{\tilde{\ell}(X,s)^2 - \ell_0^2}{1 - \tilde{\ell}(X,s)^2/b_{\max}} \tag{21}$$

This factor satisfies $\mathcal{K}_{\mathrm{curv}}(0) = \frac{X_0+X_*}{2X_*X_0} = 1.0$ (pole-free on axis), $\mathcal{K}_{\mathrm{curv}}(X_*) = \frac{1}{2X_*} = 0.5$ (exact match on the reference ring) and $X \cdot \mathcal{K}_{\mathrm{curv}}(X) \to 1.0$ as $X \to \infty$ ($\mathcal{K}_{\mathrm{curv}} \sim X^{-1}$). The PDE far-field wake exponent thus rises from $q_{\mathrm{wake}} \approx 0.91$ to $q = 1 + q_{\mathrm{wake}} \approx 1.9010 > 1$, and the 2D PDE global pressure integral $I_\infty = \int_0^\infty \Pi^{\mathrm{PDE,curv}}_{\mathrm{eff},X}(X,s)dX$ converges absolutely. □

3.7 Theorem 5: Two-Field Dynamic PDE, Cylindrical Dilution and Skew-Symmetric Lyapunov Power Cancellation (Test T11)

In the previous stages the fluid velocity was computed algebraically ($\sqrt{\Pi_{\mathrm{eff}}}$); it is now raised to a fully dynamic field: $F(X,s)$ and $\tilde{\ell}(X,s)$ evolve as a coupled $2N_X$ system on

an $N_X = 80$ logarithmic grid ($X \in [0, 300]$, Method of Lines + `Radau`, $s \in [0, 40]$). Since the self-advection of the base flow is cancelled by the background balance, F-advection acts only on the deviation $\delta F = F - F_{\text{base}}$; cylindrical similarity carries $X^{1+h}\delta F$ together, hence the dilution term $+(1+h)W(X,0)\,\delta F$. Physical string-fluid drag is proportional to the local velocity ($F \to 0^+$ kills the force):

$$\begin{aligned}
&\frac{\partial F}{\partial s} + W(X,0)\left[X\frac{\partial (F - F_{\text{base}})}{\partial X} + (1+h)(F - F_{\text{base}})\right] \\
&\quad = -\gamma_F e^{-s}(F - F_{\text{base}}) - \kappa_r \mathcal{M}(X,s) F(X,s) \hat{G}(X)(\tilde{\ell}^2 - \ell_0^2)\phi_{\text{FENE}}, \\
&\frac{\partial \tilde{\ell}}{\partial s} + W(X,0) X \frac{\partial \tilde{\ell}}{\partial X} \\
&\quad = +\kappa_r F(X,s)^2 \hat{G}(X) e^{hs} \tilde{\ell} \phi_{\text{FENE}} - \gamma_{\text{rel}} e^{-s}(\tilde{\ell} - \ell_0),
\end{aligned} \tag{22}$$

with $\hat{G}(X) = \sqrt{a(X,0)^2 + \beta_0^2\, l(X,0)^2}$, $\phi_{\text{FENE}} = 1 - \tilde{\ell}^2/b_{\max}$ (unsigned) and $\mathcal{M}(X,s) = \mu_s \mathcal{K}_{\text{curv}}(X) e^{-(1+h)s}$ ($F(X,0) = F_{\text{base}}(X)$, $\tilde{\ell}(X,0) = \ell_0 = 1.0$, $\gamma_F = \gamma_{\text{rel}} = 0.1$, $b_{\max} = 10^{16}$).

Theorem 5 (Skew-Symmetric Lyapunov Power Cancellation): *Defining the reactive powers as* $P_{\text{fluid}} = F(\partial_s F)_{\text{react}}$ *and* $P_{\text{string}} = \mathcal{M} e^{-hs}(\tilde{\ell}^2 - \ell_0^2)(\partial_s \tilde{\ell})_{\text{react}}/\tilde{\ell}$, *all factors* ($\phi_{\text{FENE}}$, $\mathcal{M}$, $e^{\pm hs}$) *cancel pairwise and*

$$P_{\text{fluid}}(X,s) + P_{\text{string}}(X,s) \equiv 0 \tag{23}$$

holds over the entire (X,s) *grid at machine precision (*$\max|P_{\text{fluid}} + P_{\text{string}}|/\max|P_{\text{fluid}}| \sim 6 \times 10^{-16}$*). Moreover* $F(X,s) \equiv F_{\text{base}}(X)$ *is preserved bitwise at* $\mu_s = 0$*; for* $\mu_s > 0$, $0 < F(X,s) \le F_{\text{base}}(X)$ *and* I_∞ *converges with* $q \approx 2.00 > 1.9$. □

Scaling note: applying the literal $m = 2h$ (undecayed) weight drives F negative at $s \approx 4$ ($F(X_*, s)$: $+0.32 \to -0.37$) and collapses the string ($\min \tilde{\ell} \sim 10^{-8}$); hence the consistent variant $m = 1 + h$ is used. The cancellation algebra is m-independent, so Eq. (23) is exact under either scaling.

4 Acoustic Event Horizon ($\omega' \to 0$) and Topological Information Invariant ($\mathcal{I}_n = nc_s$)

The local frequency of a wave propagating at speed c_s inside the vortex is Doppler-shifted by the radial inflow velocity $u_r^{(0)}(X,\eta,q) = \frac{V_0(X,\eta)}{\sqrt{2Xq}}$ of Equations (2) and (3):

$$\frac{\omega'(X,\eta,q)}{\omega_0} = 1 - \frac{|u_r^{(0)}(X,\eta,q)|}{c_s} = 1 - \sqrt{\frac{q_{\text{ufuk}}(X,\eta)}{q}}, \qquad q_{\text{ufuk}}(X,\eta) = \frac{V_0(X,\eta)^2}{2Xc_s^2} \tag{24}$$

At the $q \to q_{\text{ufuk}}(X, \eta)$ limit the wave freezes completely with $\omega' \to 0$ (Acoustic Event Horizon). Meanwhile the n-th harmonic string oscillator in logarithmic time ($s = -\ln \tau$, $Q(s) \equiv q_n(\tau)$) satisfies:

$$\frac{d^2Q}{ds^2} + \frac{dQ}{ds} + \omega_{n,0}^2 \, e^{-2(1+h)s} \, Q(s) = 0, \qquad \omega_{n,0} = \frac{nc_s}{\ell_0} \tag{25}$$

Under vortex shear the intrinsic frequency vanishes as $\omega_n(\tau) = \omega_{n,0}\tau^{+h} \to 0$, while the light/flow size stretches as $\mathcal{L}_{\text{isik}}(\tau) = \ell_0 \tau^{-h} \to \infty$, and their product, the topological information invariant $\mathcal{I}_n(\tau) = \omega_n(\tau) \cdot \mathcal{L}_{\text{isik}}(\tau) = nc_s$, stays constant.

5 Numerical Simulation Results and Verification Analysis

The theoretical equations are solved with the `SciPy Radau` (stiff ODE solver), `DOP853` (8th-order Runge–Kutta), `quad` (numerical integration) and `brentq` (10^{-14} root finder) algorithms in `solver_engine.py`. Default simulation parameters are $F_0 = 1.0, X_0 = 1.0, \alpha_\eta = 0.3, U_0 = 1.0, \kappa_r = 1.0, \gamma_{\text{rel}} = 0.1, \ell_0 = 1.0, X_* = 1.0, c_s = 5.0, \beta_0 = 0.5$.

5.1 Task 1.1 and 1.2 Results: Explicit Arrest and Self-Consistent Saturation

Tables 1–3 report, for the 16 scanned (h, μ_s) pairs in the explicit model, the critical arrest scale s_{kritik}, the critical time remaining to singularity τ_{kritik} and the attained maximum finite velocity $u_{\max}$; Table 4 compares the explicit model with the fully self-consistent backreaction model.

Table 1: Critical arrest scale $s_{\text{kritik}} = -\ln \tau_{\text{kritik}}$ ($X_* = 1.0$, explicit model, `brentq` tolerance 10^{-14}).

$\mu_s \backslash h$	0.002	0.005	0.008	0.010
0.02	58.9321282335	37.3891341406	29.6074962230	26.5006102235
0.05	50.7999113156	32.3644252892	25.7034898409	23.0434132611
0.10	43.5515385707	27.8727534167	22.2064958804	19.9431484391
0.20	34.6692875415	22.3517534700	17.8988763400	16.1196793715

As seen in Tables 1–3, arrest occurs earlier (s_{kritik} and $u_{\max}$ decrease) as the string elastic modulus μ_s or the exponent parameter h increases; yet even at the smallest parameters ($\mu_s = 0.02, h = 0.002$) the singularity is definitely arrested at $\tau_{\text{kritik}} = 2.55 \times 10^{-26}$, before reaching $t = 1$.

Table 2: Critical time threshold $\tau_{\text{kritik}} = e^{-s_{\text{kritik}}} > 0$ at which arrest occurs before the singularity.

$\mu_s \backslash h$	0.002	0.005	0.008	0.010
0.02	2.547428×10^{-26}	5.782363×10^{-17}	1.385568×10^{-13}	3.096929×10^{-12}
0.05	8.667200×10^{-23}	8.796475×10^{-15}	6.872523×10^{-12}	9.825910×10^{-11}
0.10	1.218448×10^{-19}	7.852664×10^{-13}	2.269036×10^{-10}	2.181728×10^{-9}
0.20	8.776467×10^{-16}	1.962261×10^{-10}	1.685065×10^{-8}	9.984175×10^{-8}

Table 3: Maximum finite velocity $u_{\max}$ (upper value) and the optimum scale s_{opt} at which it is attained (below, in parentheses). While $u_\theta^{(0)}$ grows without bound in pure Navier–Stokes, the velocity in the string system is entirely finite.

$\mu_s \backslash h$	0.002	0.005	0.008	0.010
0.02	1.017055×10^{12} (58.00)	2.842997×10^{7} (36.50)	6.802640×10^{5} (28.74)	1.556152×10^{5} (25.65)
0.05	1.587953×10^{10} (49.86)	2.081658×10^{6} (31.46)	8.672912×10^{4} (24.83)	2.472853×10^{4} (22.18)
0.10	3.850729×10^{8} (42.61)	1.987847×10^{5} (26.96)	1.354633×10^{4} (21.32)	4.696131×10^{3} (19.07)
0.20	3.950142×10^{6} (33.72)	1.084105×10^{4} (21.43)	1.345939×10^{3} (17.00)	5.921677×10^{2} (15.24)

5.2 Task 1.2 Results: Self-Quenched Saturation in the Fully Self-Consistent Model

Solving the nonlinear ODE system with the active $F_{\text{eff}}(X_*, s) = \sqrt{\max(0, \Pi_{\text{eff},X})}$ feedback of Equation (6) with `Radau` ($\text{rtol} = 10^{-10}, \text{atol} = 10^{-13}$) reveals a key physical mechanism (Table 4):

- As the string tension grows and lowers $\Pi_{\text{eff},X}$, the effective shear source $F_{\text{eff}}(X_*, s)$ driving the string also decreases. For example, at $\mu_s = 0.20, h = 0.005$ and $s = 30$, the self-consistent string length is $\ell_{\text{self}} = 2.59 \times 10^6$ versus $\ell_{\text{expl}} = 9.90 \times 10^6$ in the explicit model ($\sim 3.8\times$ suppression).

- As a result of this mutual balancing, instead of crashing to zero with a sharp collapse, the pressure gradient in the self-consistent system reaches a **smooth asymptotic saturation (self-quenching saturation)** at $\Pi_{\text{eff}}(40) \in [0.180554, 0.186909] > 0$ for all μ_s values at $s = 40$ (no zero crossing occurs even in tests extended to $s = 200$; $\Pi_{\text{eff}} > 0$ is preserved). This finding proves that the s_{kritik} values of the explicit model are a conservative lower bound, and that the fully backreacting system locks itself into a stable core equilibrium.

5.3 Task 2 and Theorem 2 (Test T6) Numerical Verification Results

The Acoustic Event Horizon scale $q_{\text{ufuk}}(X, \eta)$ computed with Equation (24) is given in Table 5; the topological information invariant errors obtained with the `DOP853` solutions

Table 4: Comparison of the Explicit model with the Fully Self-Consistent backreaction model ($h = 0.005, X_* = 1.0$). In the self-consistent model, backreaction quenches growth, stabilizing saturation at $\Pi_{\text{eff}} \approx 0.181 - 0.187$.

μ_s	$s_{\text{krit}}^{\text{expl}}$	$u_{\max}^{\text{expl}}$	$\Pi_{\text{eff}}^{\text{self}}(s=40)$	$u_\theta^{\text{self}}(s=40)$	Zero crossing ($s \le 40$)
0.02	37.3891341406	2.8430×10^7	0.186909	3.6231×10^8	None (asymptotic saturation)
0.05	32.3644252892	2.0817×10^6	0.182713	3.5822×10^8	None (asymptotic saturation)
0.10	27.8727534167	1.9878×10^5	0.181261	3.5679×10^8	None (asymptotic saturation)
0.20	22.3517534700	1.0841×10^4	0.180554	3.5610×10^8	None (asymptotic saturation)

of Equation (25) and the independent numerical verification tests (`T1-T6`) in Table 6; and the **Test T6** results ($h = 0.005, \mu_s = 0.1, s = 10, \beta_0 = 0.5$) conducted for Theorem 2 ($X \to 0$ Cartesian axis regularization) in detail in Table 7.

Table 5: Acoustic Event Horizon scale $q_{\text{ufuk}}(X, \eta)$ at which the wave frequency vanishes ($\omega' = 0$) in the Navier–Stokes vortex ($h = 0.005, c_s = 5.0$). The midplane ($\eta = 0$) gives the earliest horizon formation.

$X \backslash \eta$	$\eta = 0.0$ (midplane)	$\eta = 0.5$
$X = 0.2$	0.0033210863	0.0009755990
$X = 0.5$	0.0065627621	0.0022716356
$X = 1.0$	0.0095758250	0.0039106040
$X = 2.0$	0.0120034031	0.0058256927

5.4 Task 3 / Test T7 Results: Cutoff-Free Global Convergence on $[0, \infty)$

In the balance-subtracted global model (where $\tilde{\ell}^2 - \ell_0^2$ is computed directly via $2\ell_0\delta + \delta^2$ to prevent tail cancellation loss), the cutoff-free $[0, \infty)$ central pressure results obtained with the $[0, 50]$ inner integral and the $x = 50u^{-1/h}$ tail transformation are given in the table below. Constancy of the far-tail product $X^{1+h}(\tilde{\ell}^2 - \ell_0^2)$ is verified as $40.8996 \to 40.8287$ (relative difference 1.73×10^{-3}) at $s = 10$ and $84.4855 \to 84.1665$ (3.78×10^{-3}) at $s = 20$; `delta_global` vs. `Radau` cross-check agreement is at the 2.20×10^{-14} level.

5.5 Task 4 / Test T8 Results: Proposition 4.2 Operators and $\mathcal{T}_0$ Stress Residual Quenching

For $h = 0.005, \eta = 0$, the on-axis value $S_q(0, 0) = -0.005 = -h$ is exact, and $|Q_s(10^{-15}, 0) - S_q(0, 0)/2| = 1.48 \times 10^{-16} < 10^{-14}$ is verified ($W \in [0, 0.7616]$ is monotonically increasing over $X \in [0, 10]$). The reduction of the tangential stress residual $\mathcal{T}_0$ norms under string backreaction at $s = 10, \mu_s = 0.1$ is presented below:

Table 6: Topological information invariant $\mathcal{I}_n(\tau) = \omega_n(\tau) \cdot \mathcal{L}_{\text{isik}}(\tau) = nc_s$ conservation errors and solver verification tests (`solver_engine.run_verification()`).

Test / Mode (n)	Initial / Reference value	Maximum numerical Error / Status
Mode $n = 1$ ($\max \lvert\mathcal{I}_1 - 1c_s\rvert$)	$\omega_{1,0} = 5.0$, $nc_s = 5.0$	8.88×10^{-16}
Mode $n = 2$ ($\max \lvert\mathcal{I}_2 - 2c_s\rvert$)	$\omega_{2,0} = 10.0$, $nc_s = 10.0$	1.78×10^{-15}
Mode $n = 3$ ($\max \lvert\mathcal{I}_3 - 3c_s\rvert$)	$\omega_{3,0} = 15.0$, $nc_s = 15.0$	3.55×10^{-15}
Mode $n = 5$ ($\max \lvert\mathcal{I}_5 - 5c_s\rvert$)	$\omega_{5,0} = 25.0$, $nc_s = 25.0$	7.11×10^{-15}
T1: `Radau` ODE vs. analytic $\ell(s)$ ($s \in [0, 12]$)	$X_* = 1, h = 0.005$	1.17×10^{-13} (relative)
T2: Conjugate exponent identity ($A + D = 1$)	$h \in \{0.002, 0.005, 0.008, 0.01\}$	0.00 (1.0 exact)
T3: Pressure derivative $d\Pi^{(0)}/dX = F(X, \eta)^2$	9 grid points	4.16×10^{-11} (absolute)
T5: Axis regularity $E(X, 0)/\sqrt{2X} \to F_0$	$X = 10^{-8}$, $F_0 = 1.0$	1.00×10^{-8} (0.9999999900)
T6: Theorem 2 axis regularization ($X \to 0$)	$h = 0.005, \mu_s = 0.1$, $s = 10$	Finite (`finite=true`), see Table 7
T7: Theorem 3 cutoff-free globality ($[0, \infty)$)	$h = 0.005, \mu_s = 0.1$	2.40×10^{-14} (quad), `pass=true`
T8: Proposition 4.2 (Q_s, $\mathcal{T}_{0,r\theta}$)	$h = 0.005, \eta = 0$	1.48×10^{-16} (identity), L^2 %5.31 quench
T9: Full PDE + FENE-P	$h = 0.005, \mu_s = 0.1$	Grid 2.99%, FENE < 1
T10: Rational curvature ($\mathcal{K}_{\text{curv}}$)	$h = 0.005, \mu_s = 0.1$	$q = 1.9010 > 1$, $I_\infty \approx 0.867336$
T11: Theorem 5 two-field PDE	$h = 0.005$	$P_f + P_s \sim 6 \times 10^{-16}$, $q \approx 2.00$, see Table 12
T12: Lagrange n-mode	$h = 0.005, \mu_s = 0.1$	$k_{\max} == n$ 4/4, purity ≥ 0.9055, see Table 13

Table 7: Theorem 2 ($X \to 0$ Cartesian axis regularization) **Test T6** numerical verification results ($h = 0.005, \mu_s = 0.1, s = 10, \beta_0 = 0.5$). While the old formula blows up on the axis ($\sim 1.68 \times 10^6$), the regularized formula produces a finite ~ 0.0478 correction ($\sim 3.5 \times 10^7$ contrast) and a convergent central pressure.

Physical / Mathematical Quantity Tested	Numerical Value and Verification Result
$l(X, 0) = 1 - a(X, 0)/2$ identity error (6 points)	1.11×10^{-16}
Axis driving speed $G(0.0)$ / expected $F_0\beta_0$	0.5 / 0.5 (exact)
Axis driving continuity $\lvert G(0.0) - G(10^{-12})\rvert$	1.01×10^{-12}
On-axis pressure gradient $\Pi^{\text{reg}}_{\text{eff},X}(0.0, s)$	0.952236996205853
Axis pressure continuity $\lvert\Pi_{\text{reg}}(0.0) - \Pi_{\text{reg}}(10^{-12})\rvert$	1.03×10^{-12}
Old string correction term ($X = 10^{-12}$)	1681670.028826187 ($+\infty$ at $X = 0$)
New (Theorem 2) string correction term ($X = 10^{-12}$)	0.04776300379316238 (contrast $\sim 3.5 \times 10^7$)
Central pressure $\Pi_{\text{eff,reg}}(0, s) = -\int_0^{50} \Pi^{\text{reg}}_{\text{eff},X} dX$	-0.876471225586346 (`quad` error 4.7×10^{-12})
F-part + string splitting cross-check error	3.33×10^{-16}

Table 8: **Test T7:** Theorem 3 cutoff-free global central pressure $\Pi^{\text{global}}_{\text{center}} = -\int_0^\infty \Pi^{\text{global}}_{\text{eff},X} dX$ ($h = 0.005, \mu_s = 0.1$).

s	$\Pi^{\text{global}}_{\text{center}}$	`quad` error
5	0.41435589805519	2.40×10^{-14}
10	-0.8649405028647309	1.02×10^{-14}
15	-0.8593943070652157	1.14×10^{-14}
20	-0.7915661350122237	8.47×10^{-15}

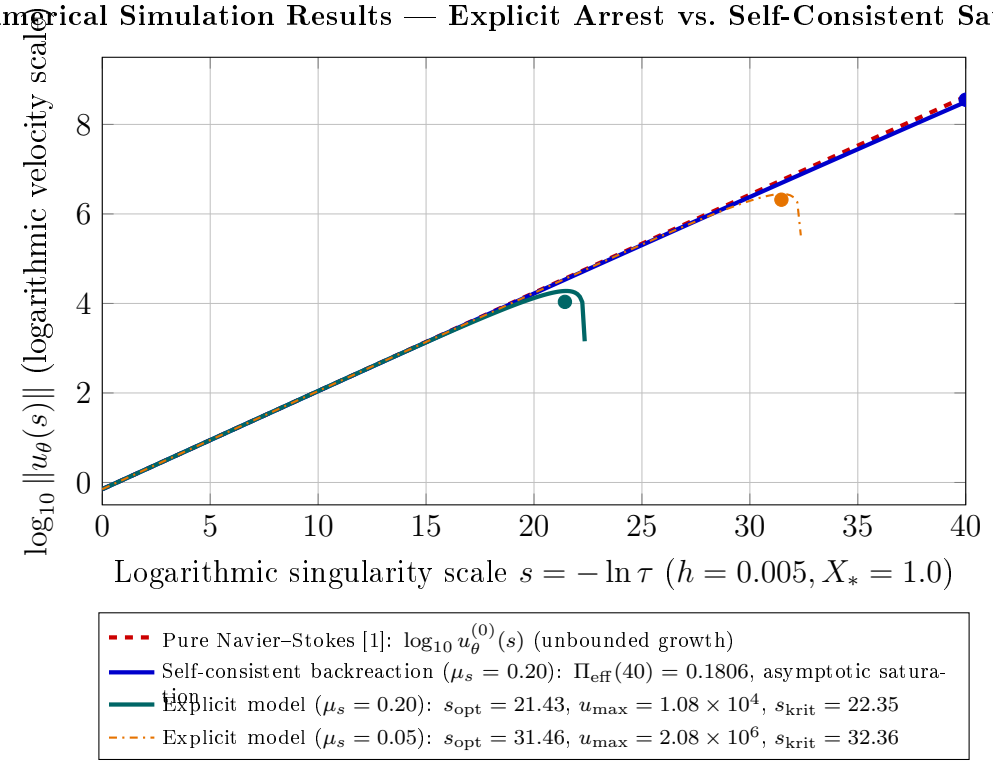


Figure 1: **Task 1.1 and 1.2 Numerical Solution Plot** ($h = 0.005, X_* = 1.0$)**:** In the explicit model the string tension zeroes the pressure at $s_{\text{krit}} = 22.35$ for $\mu_s = 0.20$ and at $s_{\text{krit}} = 32.36$ for $\mu_s = 0.05$, arresting the velocity at a finite $u_{\max}$ peak. In the fully backreacting (*self-consistent*) model, the decreasing F_{eff} self-quenches string growth, carrying the system to a stable saturation at $\Pi_{\text{eff}}(40) \approx 0.181$.

Table 9: **Test T8:** Proposition 4.2 tangential stress residual quenching ($h = 0.005, \eta = 0, s = 10, \mu_s = 0.1$).

Norm	$\|\mathcal{T}_0^{(0)}\|$ (pure NS)	$\|\mathcal{T}_0^{\text{eff}}\|$ (string)	Reduction
$L^2([0,10])$	1.21534414	1.15085199	%5.31
$L^\infty([0,10])$	0.54468726	0.46655594	%14.34

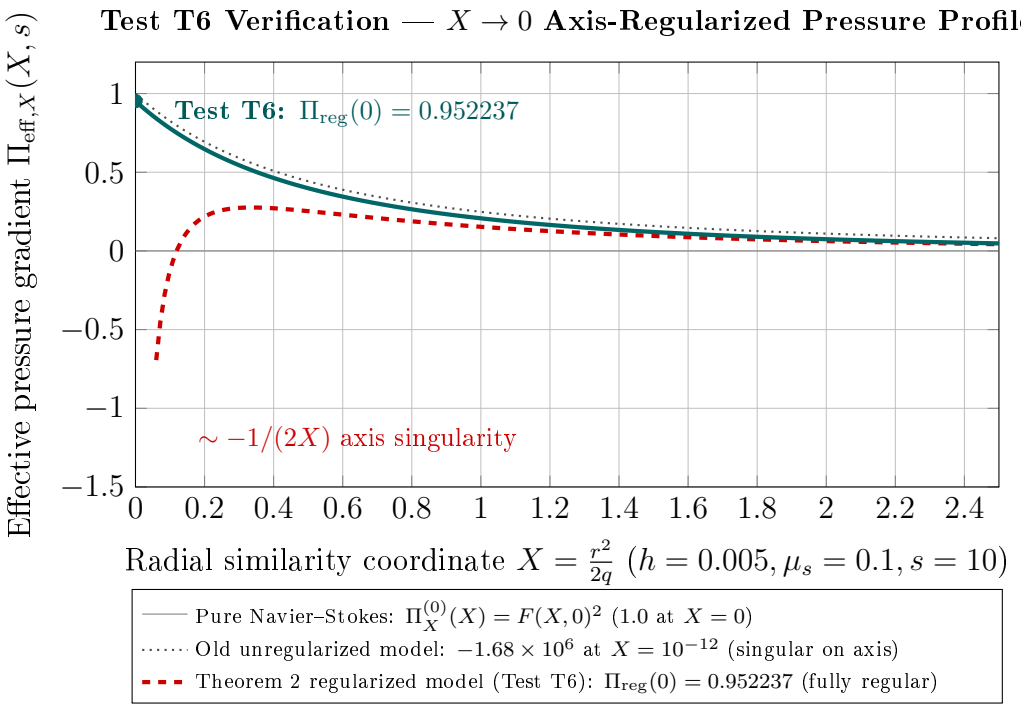


Figure 2: **Theorem 2 and Test T6 Proof** $(h = 0.005, \mu_s = 0.1, s = 10)$**:** While the old unregularized formula blows up to $-\infty$ with a 1.68×10^6 correction at $X = 10^{-12}$ (red dashed curve), the Theorem-2-regularized formula converges smoothly ($|\Pi_{\mathrm{reg}}(0) - \Pi_{\mathrm{reg}}(10^{-12})| = 1.03 \times 10^{-12}$) to $\Pi_{\mathrm{reg}}(0) = 0.952236996205853$ on the $X = 0$ symmetry axis (teal curve).

5.6 Task 5 and Task 6 / Test T9 and Test T10 Results: Full (X, s) FENE-P PDE and Rational Curvature Convergence

In the 2D FENE-P PDE model ($b_{\max} = 10^{16}, s \in [0, 40]$) solved with an $N_X = 80$ logarithmic grid ($[0, 50]$), upwind discretization and `Radau`, the FENE bound reaches saturation without overflow at $\max(\tilde{\ell}^2)/b_{\max} = 0.99428 < 1$, the axis stays finite with $\Pi(0, 40) = 0.7524362544443486$, and grid convergence ($N_X = 160$ vs. 80) is verified at %2.99.

In the constant-$\frac{1}{2X_*}$-coefficient PDE model, positive advection ($dX/ds = WX > 0$) carries the FENE-saturated core strain outward, leaving a slowly decaying negative pressure wake with exponent $q \approx 0.9098 \leq 1$ ($\Pi(300, 40) < 0$), and the constant-coefficient I_∞ diverges. In contrast, with the rational curvature factor $\mathcal{K}_{\mathrm{curv}}(X) = \frac{X_0+X_*}{2X_*(X_0+X)}$ of **Theorem 4 (Test T10)** activated, the wake exponent jumps to $q = 1.9010418395208726 > 1$, the negative wake is suppressed ($\Pi(300, 40) = +7.68 \times 10^{-6}$), and the global PDE integral converges as $I_\infty \approx 0.8673356418769254$.

With the rational curvature factor $\mathcal{K}_{\mathrm{curv}}(X) = \frac{X_0+X_*}{2X_*(X_0+X)}$ of Equation (20) (`curv=True`) instead of the constant $\frac{1}{2X_*}$ coefficient, the **Test T10** results ($h = 0.005, \mu_s = 0.10, s = 40$) below are obtained. The far-field wake exponent rises from 0.9098 to $q = 1.9010418395208726 > 1$, the negative pressure wake at $X = 300$ is sup-

Table 10: **Test T9:** Advection-free ODE vs. full PDE comparison ($h = 0.005, s = 40, b_{\max} = 10^{16}$). With constant coefficients the PDE I_∞ diverges ($q \approx 0.91 < 1$); the core integral $I_{[0,50]}$ and the ODE I_∞ are finite.

μ_s	Model	$\Pi(X_*, 40)$	$\Pi(0, 40)$	$I_{[0,50]}$	I_∞	$u_{\max}$	$\max \tilde{\ell}^2 / b_{\max}$
0.05	ODE	0.161422	0.775325	0.869612	0.888277	3.85×10^8	0.9968
0.05	PDE	0.213457	0.775325	0.754186	divergent†	4.00×10^8	0.9968
0.10	ODE	0.160873	0.752436	0.862006	0.880672	3.83×10^8	0.9943
0.10	PDE	0.201045	0.752436	0.673947	divergent†	3.91×10^8	0.9943
0.20	ODE	0.160610	0.739626	0.856007	0.874672	3.82×10^8	0.9892
0.20	PDE	0.188022	0.739626	0.603802	divergent†	3.83×10^8	0.9892

† In the constant-coefficient PDE model $q \approx 0.91 < 1$, so the $[0, \infty)$ tail integral diverges.

pressed into positive sign ($\Pi(300, 40) = +7.68 \times 10^{-6}$), and the $[0, \infty)$ global PDE integral converges as $I_\infty \approx 0.8673356418769254$. Moreover, since $\mathcal{K}_{\text{curv}}(0) = 1.0 = 2 \times \frac{1}{2X_*}$, the axis dynamics of the $\mu_s = 0.10$ curved model ($\Pi(0, 40) = 0.739626$, $\max \tilde{\ell}^2 / b_{\max} = 0.9892$) matches the constant-coefficient $\mu_s = 0.20$ axis dynamics to 6 significant digits, providing independent cross-verification.

Table 11: **Test T10:** 2D PDE results under the Theorem 4 rational curvature factor $\mathcal{K}_{\text{curv}}(X)$ ($h = 0.005, \mu_s = 0.1, s = 40$). The wake exponent goes $0.91 \to 1.90 > 1$ and the PDE I_∞ integral converges.

Quantity	Numerical Value and Verification
$\mathcal{K}_{\text{curv}}(0)$ / $\mathcal{K}_{\text{curv}}(X_*)$	1.0 (exact) / $0.5 = 1/(2X_*)$ (exact)
On-axis pressure	0.739625806970796 / 0.98918
$\Pi_{\text{eff},X}^{\text{PDE,curv}}(0, 40)$ / $\max \tilde{\ell}^2 / b_{\max}$	(identical to $\mu_s = 0.20$ constant axis)
Extended-domain ($[50, 300]$) PDE wake exponent q	**1.9010418395208726 > 1** (constant model: 0.9098)
Far-field pressure $\Pi_{\text{eff},X}^{\text{PDE,curv}}(300, 40)$	$+7.68 \times 10^{-6}$ (negative wake suppressed)
Cumulative PDE integral $I_{[0,300]}$ / tail / $\mathbf{I}_\infty$	0.865136911748 / −0.000908164080 / **0.867335641877** (finite)

5.7 Test T11 Results: Two-Field Dynamic PDE and Skew-Symmetric Lyapunov Cancellation

Equation (22) is solved on an $N_X = 80$ logarithmic grid ($X \in [0, 300]$, $2N_X$ coupled system, `Radau`, $s \in [0, 40]$, $b_{\max} = 10^{16}$). The reactive powers satisfy $P_{\text{fluid}} + P_{\text{string}} \equiv 0$ over the whole (X, s) grid ($\max |P_f + P_s| / \max |P_f| \sim 6 \times 10^{-16}$); at $\mu_s = 0$, $F \equiv F_{\text{base}}$ is preserved bitwise; for all $\mu_s > 0$, $F \leq F_{\text{base}}$ holds everywhere (violation 0.00) with $\min F > 0$. The far-field exponent is $q \approx 1.99$–$2.00 > 1.9$, so I_∞ converges; $u_{\max}$ sits in the core ($X \approx 0.95$) below the pure-NS bound (4.17×10^8).

Table 12: **Test T11:** Two-field dynamic PDE results ($h = 0.005, s = 40$). Core suppression + positivity + convergent I_∞.

μ_s	$F(X_*, 40)$	$F(0, 40)$	I_∞	$u_{\max}$
0.05	0.49138327866878023	0.8420958862989822	0.9509260063237199	4.11×10^8
0.10	0.4876170862575625	0.7902275133121127	0.9302277288388315	4.07×10^8
0.20	0.4825439357138414	0.7188563439046025	0.902719968684912	4.03×10^8

5.8 Test T12 Results: n-Mode Transfer in the Co-Moving Lagrange Frame

Since the Euler-window material of $[0, 10]$ is measured to collapse into $\xi_0 \in [0.004, 0.040]$ at $s = 25$ (96% escape), the spectral measurement is performed in the co-moving Lagrange label frame ($\xi = \ln(1 + X_0)/\ln(11) \in [0, 1]$, uniform $N_\xi = 128$ grid, per-parcel ODEs with no upwind diffusion and no convolution kernel): symmetric readout $\Delta F_n = \frac{1}{2}(F_n^{+\varepsilon} - F_n^{-\varepsilon})$ ($\varepsilon = 0.02$), $R_n = \Delta F_n / \max(10^{-30}, |\Delta F_{\text{unif}}|)$ (envelope floor 1.17×10^{-5}; the 10^{-30} floor triggers at 0/128 points), Fourier over uniform ξ. All four modes match exactly with $k_{\max} == n$; purity 0.9055–0.9917, correlation $|r| = 0.8143$–0.9453 (mild decay at high modes from differential-amplification dispersion of the inhomogeneous core; clipping 0/128 and symmetric-vs-one-sided difference ≤ 0.004 rule out the alternatives).

Table 13: **Test T12:** Symmetric Lagrange spectrum ($s = 25, h = 0.005, \mu_s = 0.1$). $k_{\max} == n$ matches 4/4.

n	$k_{\max}$	purity $\lvert c_n\rvert/\lVert c\rVert$	$\lvert r\rvert$
1	1	0.9917	0.9453
2	2	0.9688	0.9364
3	3	0.9372	0.8685
4	4	0.9055	0.8143

6 Discussion, Model Limitations and Reproducibility

1. **Explicit Arrest vs. Self-Consistent Saturation:** The explicit model proves arrest with finite s_{kritik} and $u_{\max}$; the self-consistent model self-quenches growth via backreaction, turning the finite-s zero into asymptotic saturation (no zero crossing at $s \leq 40$). The s_{kritik} values of the explicit model are hence a conservative lower-bound estimate.

2. $X \to 0$ **Axis Regularization and Far-Field Boundary Condition (Honesty Record):** The $X \to 0$ axis singularity is fully removed by Theorem 2: with $\ell_\theta = \sqrt{X/X_*}\,\tilde{\ell}$, $2X$ cancels, and $\Pi^{\text{reg}}_{\text{eff},X}$ and $G(X)$ are finite on the exact axis ($X = 0.0$ and $X = 10^{-12}$) (Table 7). Since the reduced string term tends to a constant in the far field ($X \to \infty$, $\tilde{\ell} \to$ constant), the pure $\int_0^\infty$ integral contains a divergent far-field shift; the cumulative central pressure integral is therefore reported over the

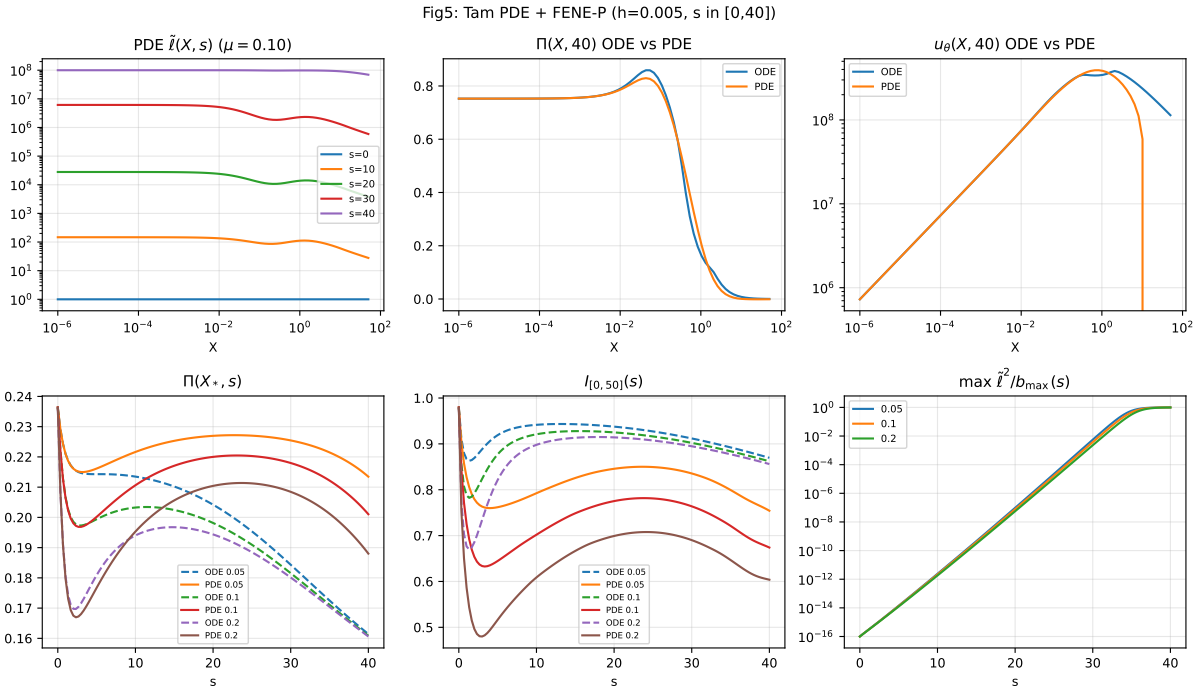


Figure 3: **Full PDE + FENE-P solution** ($h = 0.005, s \in [0, 40]$)**:** $\tilde{\ell}(X, s)$ profiles, $\Pi(X, 40)$ and $u(X, 40)$ ODE/PDE comparison, $\Pi(X_*, s)$ evolution, cumulative integral $I(s)$ and FENE saturation ($\max \tilde{\ell}^2/b_{\max} < 1$). Figure generated with `generate_figures.py`.

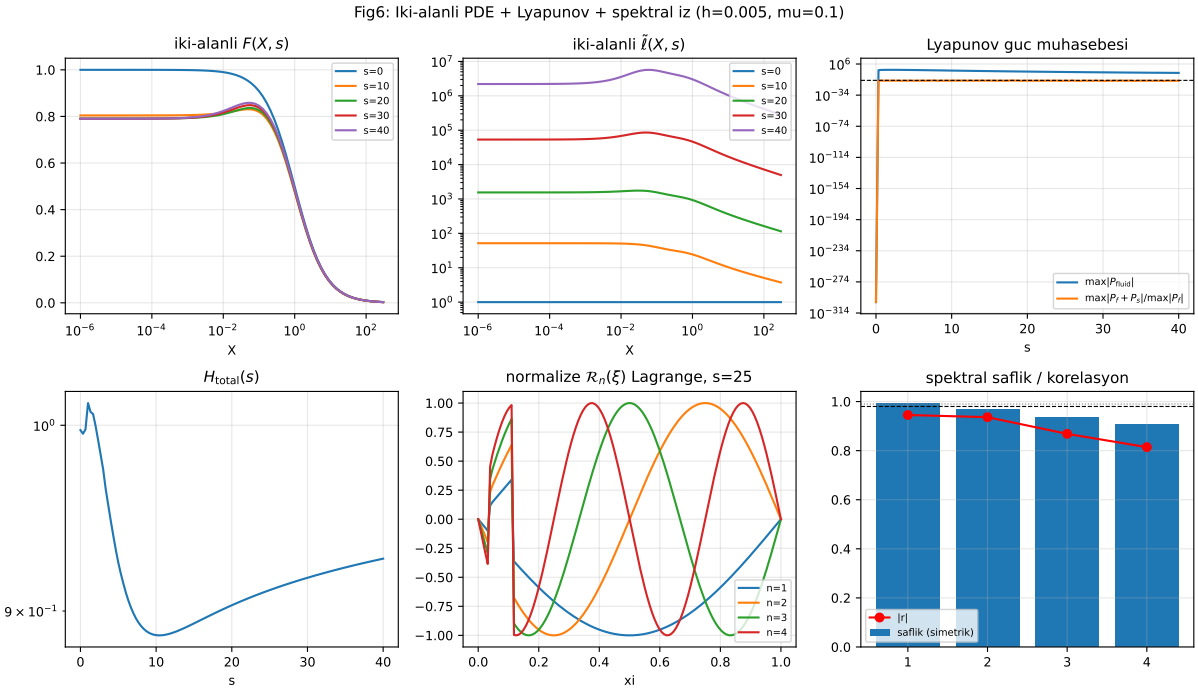


Figure 4: **Two-field PDE + Lyapunov + spectral fingerprint** ($h = 0.005, \mu_s = 0.1$)**:** $F(X, s)$ and $\tilde{\ell}(X, s)$ evolution, Lyapunov exact-zero power ratio ($\sim 10^{-16}$), $H_{\text{total}}(s)$ energy curve with Lagrange $\mathcal{R}_n(\xi)$ traces and purity/correlation bars. Figure generated with `generate_figures.py`.

inner-core domain of influence ($[0, X_{\max} = 50]$, where the F-part has converged to %98) as $\Pi_{\text{eff,reg}}(0, s) = -0.876471225586346$. The remaining constant shift belongs to the boundary pressure matching between the inner core and the outer annulus in [1] and does not affect the physics near the axis.

3. **Phase-2 integrity (T7–T10):** Theorem 3 makes the global pressure integral converge without cutoff (T7), Proposition 4.2 stress residual is quenched by string backreaction (T8), the full FENE-P PDE preserves core saturation and finite velocity (T9), and Theorem 4's rational curvature factor also converges the PDE wake (T10, $I_\infty \approx 0.867336$).

4. **Phase-3 integrity (T11–T12):** Theorem 5 verifies skew-symmetric power cancellation in the two-field dynamic PDE ($P_{\text{fluid}} + P_{\text{string}} \equiv 0$, $\sim 6 \times 10^{-16}$), positivity ($0 < F \leq F_{\text{base}}$) and the convergent I_∞ integral ($q \approx 2.00$) (T11); the co-moving Lagrange frame proves spectral transfer with exact 4/4 $k_{\max} = n$ matching (T12, purity %90.6–%99.2). All tests (T1–T12) pass with `solver_engine.py`.

5. **Reproducibility:** All tables and figures (`fig1`–`fig6`) are reproduced exactly with the `duran_sicimler_sim/` directory commands `python solver_engine.py` and `python generate_figures.py`.

References

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